

Unit II – Empathy and Problem Definition

Detailed lecture notes — users and stakeholders, empathy mapping, observation in the field, interviewing and need finding, and the framing of a problem statement that a team can design against.

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SYLLABUS

UNIT II — EMPATHY AND PROBLEM DEFINITION (6 periods)

Understanding users and stakeholders. Empathy mapping. Observation techniques. Interviews and need finding. Problem identification and framing problem statements. Point of view statements and problem reframing.

Course Outcome CO2: Identify user needs and formulate problem statements using empathy-based approaches.

Text: Tim Brown, *Change by Design*, Harper Collins, 2012; Idris Mootee, *Design Thinking for Strategic Innovation*, 1st Edn., John Wiley & Sons, 2013.

References: Jeanne Liedtka and Tim Ogilvie, *Designing for Growth*, Columbia Business School Publishing, 2011 — the toolkit chapters on visualisation, journey mapping and mind mapping.

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Learning objectives. After studying this unit you should be able to: (i) distinguish users, customers, gatekeepers and influencers, and explain why extreme users are studied; (ii) construct a

stakeholder map on a power–interest grid and state the action appropriate to each quadrant; (iii) explain what empathy means as a design method and describe the say–do gap; (iv) build and read an empathy map from field notes; (v) select an appropriate observation technique and record observations using the AEIOU framework, separating observation from interpretation; (vi) design an interview guide, apply laddering and the five whys, and distinguish stated, observed and latent needs; (vii) synthesise field notes into insights and write a problem statement containing a user, a need and an insight; and (viii) write and test a point-of-view statement and reframe a problem deliberately.

1. Introduction – why the problem must be found before it is solved

Unit I established that real problems arrive as situations rather than as questions, and that the framing of the problem decides whether everything that follows is worth doing. This unit is about the work of that framing. It covers the first two stages of the design thinking process – **Empathise** and **Define** – in the detail needed to carry them out rather than merely to describe them.

The sequence is deliberate. You cannot frame a problem you have not investigated, and you cannot investigate a problem without deciding whom to investigate it with. So the unit begins with people – who they are and how they relate to the problem – then moves to the methods of finding out what they need, and ends with the writing of a problem statement precise enough for a team to design against.

KEY POINT The output of this unit is one sentence: a point-of-view statement naming a user, a need and an insight. Everything in the six hours exists to make that sentence defensible.

2. Understanding users and stakeholders

2.1 The user is not you

The first discipline of user research is the recognition that the designer is almost never the user. You know too much: you designed the thing, so you can always find the button. You are also not typical – your age, education, income, language and confidence with technology are all unusual by comparison with the general population. Designing for yourself is legitimate only when you are genuinely the user and can demonstrate it.

It follows that every belief you hold about a user is a **hypothesis** until somebody checks it. The practical discipline is to write those beliefs down before going into the field, so that the field work can contradict them. A list of assumptions that was never written down cannot be shown to have been wrong.

2.2 Four roles in every project

Role	What they do	Why they are easy to miss
User	Uses the thing and lives with every design decision	Often has no voice in the purchasing decision
Customer	Pays for it; frequently never touches it	Their criteria are cost and compliance, not experience
Gatekeeper	Can permit or forbid its use	Appears only at the end, and can stop the project outright
Influencer	Shapes the decision without making it	Has no formal role, so never appears on an organisation chart

2.3 Mainstream and extreme users

Users are not uniform. Most fall in the middle of any range of ability, experience or need; a minority sit at the extremes. **Extreme users** – the expert who pushes a product past its intended use and the novice who cannot begin – meet the weaknesses of a design sooner and harder than average users do, and they have usually already invented workarounds. A workaround is a pain and a design brief at the same time: somebody has already built you a prototype.

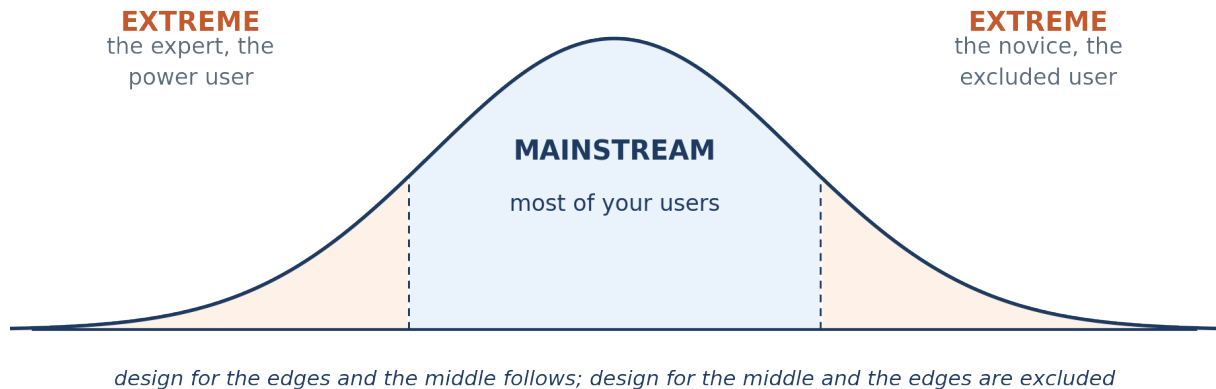


Figure 2.1 Mainstream and extreme users. Designing for the edges usually serves the middle as well; the reverse is not true.

KEY POINT Interview three extreme users before you interview thirty average ones. What an extreme user needs is usually what everybody needs, expressed more visibly.

3. Stakeholder mapping and the power–interest grid

A stakeholder map lists everybody affected by the problem, not only the person in front of you. It shows who must be consulted, who must be informed and who can simply refuse; it surfaces conflicts early, when different groups want incompatible things; and it tells you whom to interview, and in what order.

3.1 The power–interest grid

The standard tool places each stakeholder on two axes: how much **power** they have over the outcome, and how much **interest** they have in it. The action appropriate to a stakeholder follows from the quadrant they occupy.



Figure 2.2 The power–interest grid. Place each stakeholder once, then decide what each quadrant needs from the team.

Quadrant	Who they are	What you do
Manage closely	High power, high interest	Involve in framing, ideation and testing
Keep satisfied	High power, low interest	Consult before decisions; keep communication brief
Keep informed	Low power, high interest	Share progress often – these are usually your users
Monitor	Low power, low interest	Watch for change; spend no effort yet

The quadrant that repays the most attention is **low power, high interest**. This is where users normally sit, and it is precisely why user needs are so often overruled: the people who care most about the outcome have the least authority over it. Part of the designer's job is to carry their evidence into the rooms where they are not present.

3.2 Building a map in four steps

1. List every person and group the problem touches. Go wide before judging; a name on the list costs nothing.
2. Place each on power and interest, and argue the placement out loud. The argument is more valuable than the final position.
3. Mark who is missing – who is affected but has no voice in the room at all.
4. Decide one action for each quadrant and record who is responsible for it.

KEY POINT Projects fail far more often on a stakeholder nobody mapped than on a technical problem nobody solved. The gatekeeper who appears in the last week is the classic example.

IN CLASS Fifteen minutes. Map at least eight stakeholders for your team's problem on a power-interest grid, and name one you had completely forgotten until the exercise forced you to list them.

4. Personas and their limits

A **persona** is a single named, described person who stands for a group of real users. It gives the team one person to design for, a shared language in discussion, a check on the vague phrase “the user wants...”, and a face to attach to the evidence.

Its limits are equally important. A persona cannot replace meeting real people; it cannot make invented detail true; it does not cover the edges of the range, so it must be used alongside extreme users rather than instead of them; and it goes out of date unless it is updated when new evidence arrives.

A persona built from field notes is a summary. A persona built from imagination is a stereotype with a photograph attached.

5. Empathy in design – evidence, not sympathy

Sympathy is feeling for somebody. **Empathy**, in the sense used in design, is understanding another person's situation well enough to act on it. It is a working method rather than an emotional state: you go, you watch, you ask, you record, you share. It produces artefacts – notes, quotations, photographs, maps

– and those artefacts are checkable, in that another member of the team should be able to read them and reach the same conclusion.

KEY POINT If your empathy work produced nothing that another person can examine, you have not done empathy work. You have had a feeling about a problem.

5.1 Three ways to gather it

Method	What you do	What it is best for
Immerse	Live the experience yourself, even briefly	Noticing what is tiring, slow or undignified about a process
Observe	Watch people in their own setting without interrupting	Habits, workarounds and the physical environment
Engage	Talk to them about specific things that actually happened	Reasons, priorities and the history behind a behaviour

All three are used, because each catches what the others miss. Immersion without observation mistakes your own experience for everyone's; observation without engagement produces behaviour with no explanation; engagement without observation produces explanations of behaviour that may never have occurred.

6. The say–do gap

People report what they intend to do, what they remember doing and what sounds reasonable to say. What they actually do is frequently different, and they are not being dishonest. Habits are invisible to the person who has them; the sequence of small adjustments by which somebody makes an awkward process work is, to them, simply how the thing is done, and it does not occur to them to mention it.

The design consequence is a rule: ask **and** watch. Where the two disagree, you have found something. A user who says the queue is fine and then leaves without a drink has told you more by leaving than by answering.

The gap between what people say and what they do is where the best design opportunities hide.

7. The empathy map

An empathy map is one page that holds everything learned about one person. It is divided into four quadrants – what the person **says**, **thinks**, **does** and **feels** – with the user at the centre. One map is made per real person; a single map for “the user” averages people who do not exist.

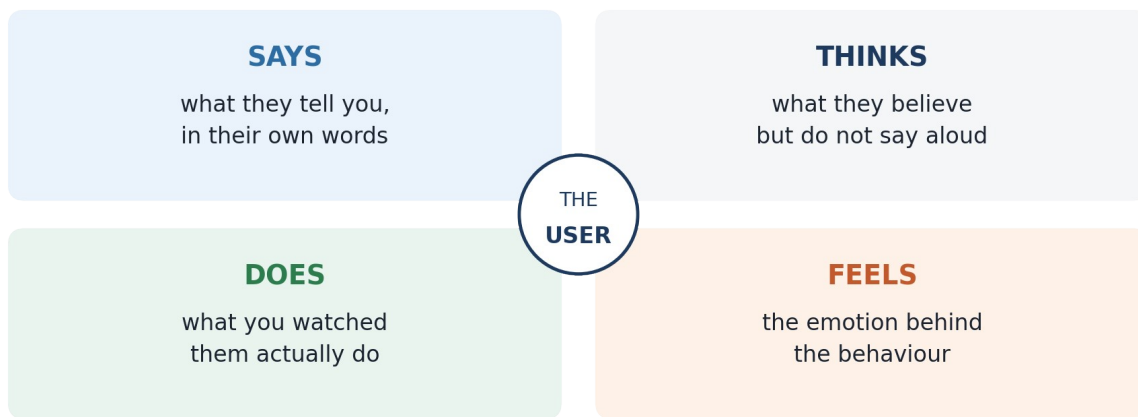


Figure 2.3 The empathy map canvas. One map per real person, filled from notes rather than from memory.

Quadrant	What goes in it	The discipline
Says	Direct quotations, in the user's own words	If it is not in quotation marks it does not belong here
Thinks	What they believe but do not say aloud	This is inferred, so mark it clearly as an inference
Does	Actions you watched them perform	Verbs only – no interpretation and no adjectives
Feels	The emotion behind the behaviour	Name it plainly: rushed, worried, embarrassed, proud

7.1 Pains and gains

Two further headings are added below or beside the quadrants. **Pains** are what frustrates or blocks the person: risks and fears, costs in time, money and effort, and the workarounds they have invented. **Gains** are what they are trying to achieve, how they measure success, what would genuinely delight them and what they already do well.

KEY POINT A workaround belongs in both columns at once. It is a pain, because the user had to invent it; and it is a design brief, because somebody has already prototyped a solution for you and tested it daily for months.

7.2 Filling one in, and reading it

- One map per person, filled straight from your notes while they are fresh.
- Use one colour for what you observed and another for what you inferred, so the two never become confused later.
- Leave the gaps visible. An empty quadrant is not a failure of the map; it tells you what to go back and find out.
- **Reading it:** look for contradictions between Says and Does – that is the say–do gap made visible on one page.
- Look for emotion with no visible cause on the map; you have missed part of the story.
- Look for the same pain appearing across several maps. One person is an anecdote; three are a pattern.

Four mistakes account for most poor empathy maps: filling them from imagination rather than from notes; writing the team's intended solution into the map as though it were a user need; making one map

for “the user” instead of one per real person; and treating the map as a form to be completed rather than as a tool to argue over.

IN CLASS Twenty minutes. Interview a classmate about their journey to college, then build one empathy map each from your own notes and compare. The differences between two maps of the same conversation are the lesson.

8. Observation techniques

Interviews give you the story people tell about themselves. Observation gives you the one they live. Four things are reliably invisible in an interview and reliably visible in the field: habits the person no longer notices; workarounds so normal to them that they never come up; the physical setting, with its noise, heat, light, clutter and interruption; and how long things actually take, as against how long they feel.

8.1 How involved should the observer be?

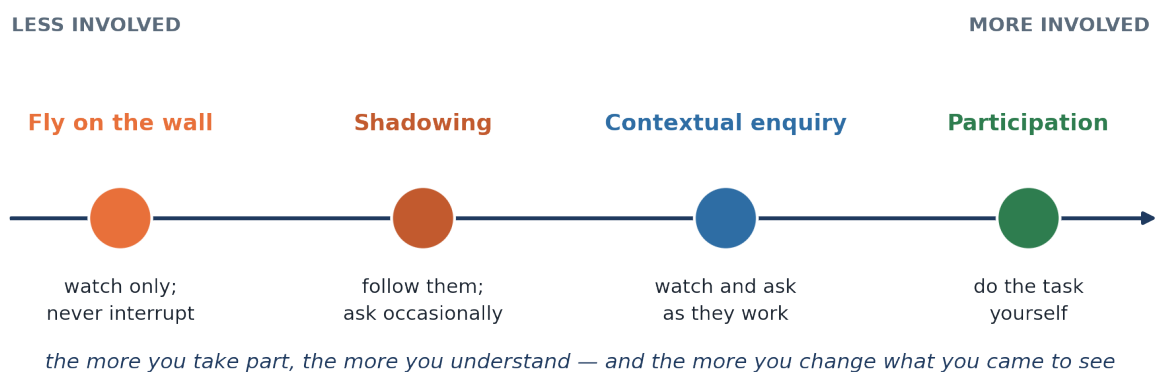


Figure 2.4 Four positions on one scale of involvement. The more you take part, the more you understand – and the more you change what you came to see.

Technique	What you do	Best used when
Fly on the wall	Sit still and watch; never interrupt	A public place, or a first visit before you know what matters
Shadowing	Follow one person through a task or a shift, asking between steps	A process that crosses several places or several people
Contextual enquiry	Watch them work and ask about it as they work	A skilled task where the reasons are not visible from outside
Participation	Do the task yourself, under supervision	When physical effort, discomfort or timing is the point

KEY POINT Your presence always changes the scene. Stay long enough for people to stop noticing you, and record what changes in the first ten minutes after you arrive – that difference is itself a finding.

9. The AEIOU framework and the field record

AEIOU is a checklist that prevents the observer from noticing only what they expected to notice. Five headings – Activities, Environment, Interactions, Objects and Users – between them catch almost everything worth recording in a field visit.

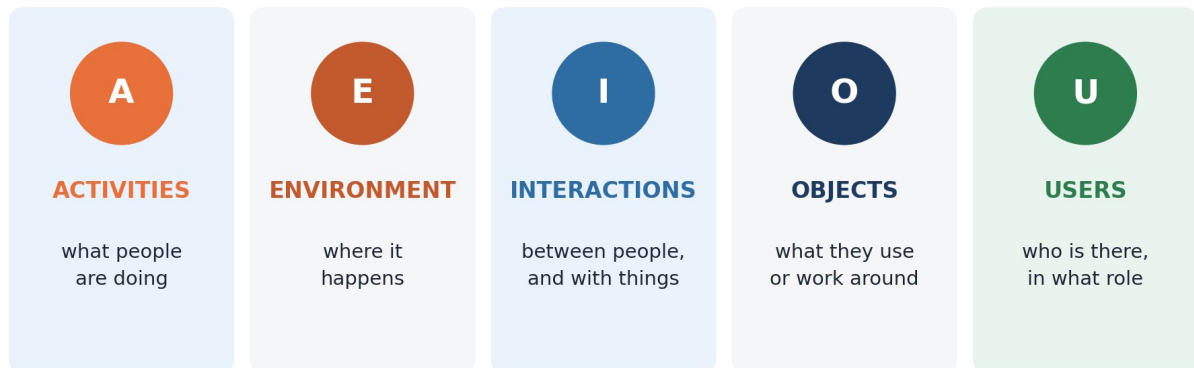


Figure 2.5 The AEIOU framework. Working through all five headings forces attention onto things a focused observer walks past.

Letter	Look for	Example in a college canteen
Activities	Goal-directed sequences of action	Queuing, paying, hunting for a seat, returning a tray
Environment	The space and its character	Noise, crowding, where the fans are, how far the bins are
Interactions	Person to person, and person to thing	Shouting an order across a counter; a token handed over
Objects	What is used, made or worked around	Trays, tokens, a chair used as a table, a wedged door
Users	Who is present and in what role	Staff, students, a visiting parent, a delivery worker

9.1 Recording what you see

- **Two columns, always.** What you observed on the left; what you think it means on the right. Mixing them destroys the evidence.
- Capture quotations word for word. A paraphrase preserves the meaning you took and discards the meaning you missed.
- Photograph the setting and the objects rather than faces, and only after asking.
- Write up within the hour. Memory decays fast, and the detail is the entire value of the visit.

“She put the tray down twice” is an observation. “She was frustrated” is an interpretation. Keep them in separate columns.

10. Ethics in the field

Field work involves real people who owe you nothing. The obligations below are not administrative formalities; they are the conditions under which the work is legitimate.

- Say who you are, what you are doing and what the notes will be used for, in plain language and before you begin.
- Anyone may decline at any moment, without giving a reason, and without being asked to reconsider.
- Never photograph or record a person without asking first. Consent to be observed is not consent to be photographed.
- Take particular care with children, patients, and anyone who depends on the service you are studying – they may feel unable to refuse.
- Store notes without names, and share findings rather than identities.

KEY POINT Consent is not a form collected once at the door. It continues through the visit, and a person who agreed at the start may withdraw at any point.

IN CLASS Twenty minutes in the field, in pairs. Pick one place on campus where people wait. Two observers, two columns, no interviewing. Record against AEIOU and count every workaround you see. Then compare: what did your partner see that you did not?

11. Interviews and need finding

Most poor interview data is created by the interviewer. Four faults account for nearly all of it: leading questions, which return the answer you expected; hypothetical questions, which return polite fiction; the desire of the person to be helpful, which makes them agree with your idea; and asking about the general case, which returns a generalisation instead of an event.

Never ask whether somebody would use your idea. Ask what they did the last time the problem actually happened.

11.1 The interview funnel

A good interview narrows. It opens wide and comfortable, moves to a specific occasion, then to the detail of that occasion, and only then to why it mattered. Reversing the order – starting with the deep question – produces either silence or an invented answer.



Figure 2.6 The interview funnel. Open wide, then narrow – never the other way round.

Ask this	Not this	Why
Tell me about the last time...	Would you use an app that...	An event can be remembered; a hypothetical cannot

Ask this	Not this	Why
Walk me through how you...	Do you think X is a problem?	A walkthrough surfaces steps the person forgot they took
Why was that difficult?	Don't you find X annoying?	The second question contains its own answer
What did you do instead?	How often do you usually...	Behaviour is specific; frequency estimates are unreliable

11.2 Writing and running an interview guide

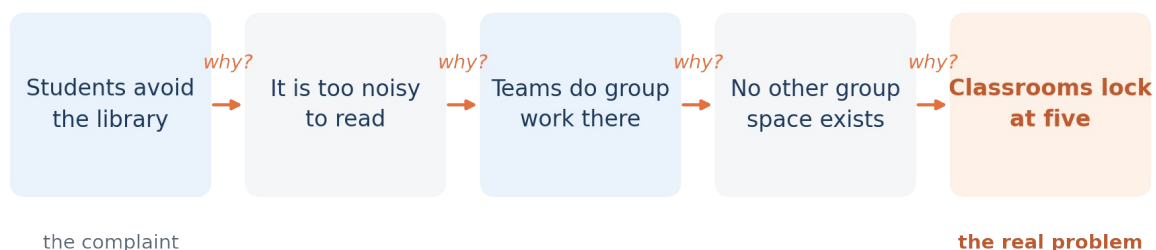
- Eight to ten questions on a single page. You will not use them all, and you should not try to.
- Open with something easy and factual, to settle the person and to establish that there are no wrong answers.
- Place the sensitive questions late, once a little trust exists.
- Leave silence after an answer. People fill silence, and what they fill it with is usually the real answer.
- Two people per interview where possible: one asks, one writes. The writer does not speak.

IN CLASS Ten minutes. Write five questions for your team's user, swap with another team, and mark every leading question you can find in theirs. Most teams find two or three.

12. Laddering, the five whys and latent needs

12.1 Getting beneath the first answer

The first answer to a question is almost never the problem; it is the complaint about the problem. **Laddering** is the technique of moving from that complaint towards a cause you can act on, by repeatedly asking why. The popular form of this is the **five whys**, though five is a guide rather than a rule: you stop when the answer stops changing, or when you reach something outside your power to influence.



five is a guide, not a rule — stop when the answer stops changing

Figure 2.7 A worked five-whys chain. Each step moves from the symptom towards something that can actually be designed.

Two cautions. Ask gently and vary the wording, or the sequence becomes an interrogation and the person begins to defend rather than explain. And ladder downwards as well as upwards: "how exactly" and "what happened then" recover the concrete detail that abstraction hides.

12.2 Three kinds of need

Kind	How you find it	Example
Stated	The person tells you directly	"I want a faster printer"
Observed	You watch them work	They print every document twice because page one jams
Latent	Inferred from the gap between the two	They need to stop having to check, not to print faster

Latent needs are needs people cannot articulate, usually because they stopped noticing the problem long ago. They are found by combining what you watched with what you were told, not by asking harder questions. A latent need is always phrased as a verb – "needs to know it worked", not "needs a status light" – and it is tested by describing it back to the person. If they say "yes, exactly", you have it.

KEY POINT Stated needs are the easiest to collect and the least valuable, because your competitors have collected them too. Latent needs are where the design opportunity is, and they are the reason the field work is worth the time it costs.

13. Synthesis – from field notes to insight

After the field work a team has pages of notes, quotations, photographs and half-formed impressions; several contradictions; at least one surprise; and no obvious answer. This is the correct state to be in. It means the team gathered evidence rather than collecting confirmation for a conclusion it had already reached.

If the field produced no surprise, you were not researching. You were gathering evidence for an answer you had already chosen.

13.1 How synthesis is done

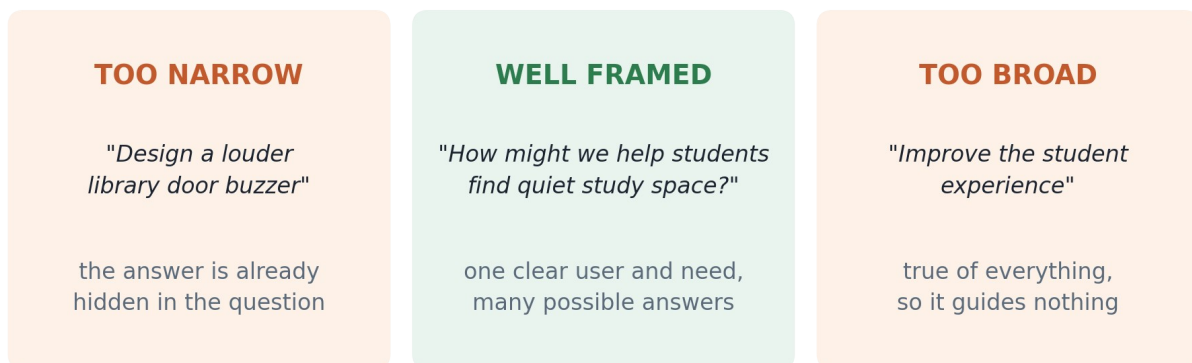
1. Write every observation on its own note – one idea per note. A note carrying three ideas cannot be clustered.
2. Cluster by what the notes have in common, not by which interview they came from. Cross-interview clusters are the ones that matter.
3. Name each cluster with a full sentence. A word is a category; a sentence is a finding, and only a finding can be argued with.
4. Identify the insight – the
5. **because**
6. behind a behaviour you did not expect.

	Observation	Insight
States	What happened	Why it happened
Form	"Four of five users refilled at a tap instead"	"They will trade water quality for time when they are late"
Test of a good one	It is checkable against the notes	It predicts what a new user would do
Use	Evidence	The engine of the problem statement

14. Framing the problem statement and How Might We questions

14.1 What makes a frame work

A problem can be framed too narrowly, too broadly, or around a solution. A frame that is too narrow contains its own answer, so there is nothing left to design; one that is too broad is true of everything and therefore guides nothing; and one that is solution-shaped names a technology instead of a need, locking the team in before it has begun.



narrow enough to act on, wide enough to allow many answers

Figure 2.8 The same situation framed three ways. Only the middle frame is designable.

Fault	Example	Why it fails
Too narrow	"Design a louder library door buzzer"	The answer is already inside the question
Too broad	"Improve the student experience"	True of everything; rules nothing out
Solution-shaped	"Build a seat-booking app"	Names a technology, not a need
Well framed	"Commuting students need somewhere to work between classes"	One user, one verb-shaped need, many possible answers

14.2 The anatomy of a problem statement

A usable problem statement has exactly three components and nothing else. It names **one user**, specifically – not "people" and not "students in general". It states **a need** as something the user must be able to **do**, not a thing they must be given. And it gives **an insight**: why the need exists and why it is not already met. It contains no solution, no technology and no brand name.

User + need + insight. If any one of the three is missing, the statement will not survive ideation.

14.3 How Might We questions

A problem statement describes a situation; a **How Might We** question turns it into an invitation to generate. The three words are chosen deliberately: **how** assumes that a way exists, **might** allows many answers rather than demanding the right one, and **we** makes it the team's work. Five or six are written from a single statement, pitched at different levels of ambition.

Too tight	Too loose	About right
How might we add a buzzer?	How might we fix education?	How might we help students find quiet space?
Only one answer fits	Any answer at all fits	Several genuinely different answers are possible
Contains the solution	Contains no user and no need	Contains a user and a need, and no solution

KEY POINT A good How Might We question makes you want to start writing ideas immediately. If it does not, it is not yet a How Might We question – it is a restatement of the topic.

IN CLASS Twenty-five minutes. Cluster your team's field notes and name each cluster in a full sentence. Write one problem statement – user, need, insight, in that order – and generate six How Might We questions from it at different scales. Pin up the statement and the six questions.

15. Point of view statements and problem reframing

15.1 The POV template

A point-of-view statement is the problem statement written in a fixed form so that its three components can be checked at a glance. The template is: **a specific user** needs **a verb-shaped need** because **an insight**.

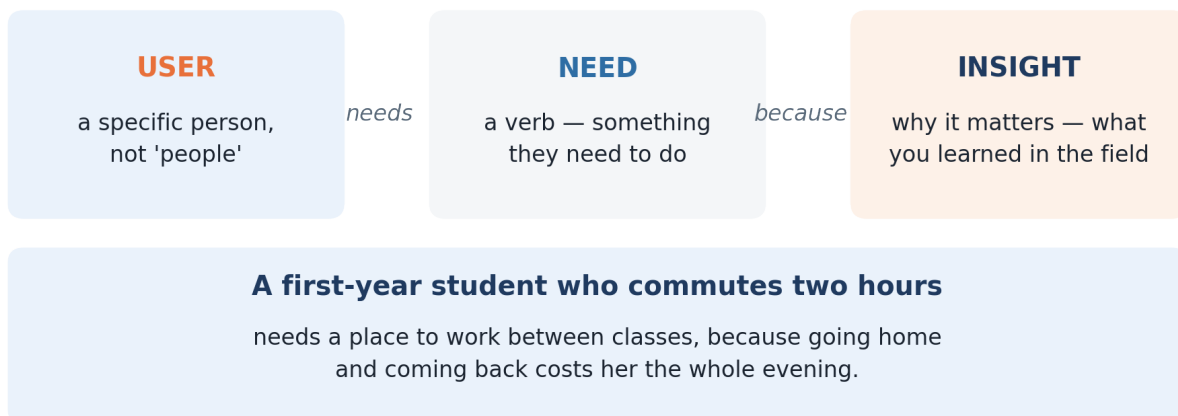


Figure 2.9 The point-of-view template with a worked example. The insight is the part that could only have been written after field work.

15.2 Needs are verbs, not nouns

“Needs a handrail” is a solution wearing the costume of a need. “Needs to feel steady on the stairs” is a need. The distinction is not pedantic: a verb keeps the solution space open, while a noun closes it before ideation has begun. The working test is whether you can name three completely different ways of meeting the need. If you cannot, you have written a solution.

The moment a noun enters the need, ideation is over before it started.

15.3 Testing a point of view

- Would the user recognise themselves in it, and agree that the need is real?
- Does it contain an insight that you did not have two weeks ago? If you could have written it before the field work, the field work has not yet been used.
- Does it suggest at least five genuinely different directions?
- Can a stranger read it and begin designing without asking you a question?

15.4 Reframing

Reframing is the deliberate restatement of a problem so that a different set of answers becomes visible. It is not a failure of the first frame; it is how the process is meant to work. It is triggered by field evidence that keeps contradicting the brief, by a prototype that fails for an unexpected reason, or by a constraint that cannot be moved and turns out to be the real subject.

Move	What you ask	Example
Zoom out	What larger problem is this a symptom of?	The queue may really be a timetabling problem
Zoom in	What if one moment were the whole problem?	Not the whole journey – just the thirty seconds at the counter
Flip it	What does the opposite reveal?	How might we make people want to wait?
Change the actor	What if somebody else did the job?	What if the user did it instead of the staff?

Every case study in Unit I turned on exactly one reframing. Embrace moved from hospitals to villages; Aravind from surgical technique to volume and cost; the Jaipur Foot from a shod life to a barefoot one; MittiCool from a cheaper refrigerator to cooling without electricity at all.

KEY POINT Reframe early, while it is cheap. Once something has been built, the cost of reframing rises steeply and the team's willingness to do it falls just as fast.

IN CLASS Twenty minutes. Write your team's point of view in one sentence on a single sheet and pass it to the next team; they write two reframes of it. Take it back, adopt one or defend the original, rewrite and pin it up. This statement is what Unit III will ideate against.

16. Summary

- You are almost never the user; every belief about a user is a hypothesis until somebody checks it, and the hypotheses should be written down before the field work.
- User, customer, gatekeeper and influencer are four distinct roles; extreme users reveal in three interviews what thirty average users conceal.
- A stakeholder map on a power–interest grid shows who to involve, consult, inform and monitor, and exposes who is affected but has no voice.
- Personas summarise evidence; built from imagination they become stereotypes.
- Empathy in design is a method producing checkable artefacts, not a feeling; it is gathered by immersing, observing and engaging.
- People report intentions and remember selectively; the gap between what they say and what they do is where the design opportunity lies.

- An empathy map holds Says, Thinks, Does and Feels for one real person, plus pains and gains; contradictions within it are the finding, not an error.
- Observation runs from fly-on-the-wall to full participation; AEIOU prevents you from noticing only what you expected.
- Field records keep observation and interpretation in separate columns, and consent continues throughout a visit rather than being collected once.
- Interviews narrow from a comfortable opening to a specific occasion; leading and hypothetical questions manufacture the data they appear to find.
- Laddering and the five whys move from complaint to cause; needs are stated, observed or latent, and latent needs carry the opportunity.
- Synthesis clusters notes across interviews and names each cluster in a sentence; the insight is the “because” behind unexpected behaviour.
- A problem statement carries one user, a verb-shaped need and an insight; How Might We questions turn it into an invitation to ideate.
- Reframing restates the problem so different answers become visible, and is done early while it is still cheap.

17. Review questions

17.1 Short-answer (two-mark) questions

1. Why is the designer almost never the user?
2. Distinguish a user from a customer and from a gatekeeper.
3. What is an extreme user? Why are extreme users interviewed first?
4. Name the two axes of a stakeholder map and the four resulting quadrants.
5. In which quadrant do users normally sit, and why does this matter?
6. State two things a persona gives a team and two things it cannot do.
7. Distinguish sympathy from empathy as the terms are used in design.
8. Name the three ways of gathering empathy.
9. What is the say–do gap? Give one example.
10. Name the four quadrants of an empathy map.
11. Why is a workaround both a pain and a design brief?
12. Name the four positions on the observation-involvement scale.
13. Expand AEIOU and state what each letter directs attention to.
14. Give one observation and one interpretation of the same event.
15. State three obligations of an observer working in the field.
16. Why should an interviewer never ask a hypothetical question?
17. What is laddering? When do you stop asking why?
18. Distinguish stated, observed and latent needs with one example each.
19. State the three components of a problem statement.
20. Write the point-of-view template in full.
21. Why must a need be written as a verb rather than as a noun?
22. Name the four ways of reframing a problem.

17.2 Long-answer (thirteen-mark) questions

1. Explain why the designer is almost never the user, and describe the four roles that must be identified in any project. Illustrate with a campus or municipal service in which the four roles are held by different people.
2. Describe stakeholder mapping on a power-interest grid. Explain the action appropriate to each quadrant and discuss why the low-power, high-interest quadrant deserves special attention from a designer.
3. Explain what empathy means as a design method. Distinguish it from sympathy, describe the three ways of gathering it, and explain why the work must produce artefacts that another person can examine.
4. Explain the say-do gap with examples, and describe how an empathy map is constructed and read. Discuss the four common faults in poorly made empathy maps.
5. Compare fly-on-the-wall observation, shadowing, contextual enquiry and participation. For each, state what it is best suited to and how the observer's presence changes what is observed.
6. Describe the AEIOU framework and explain how a field record should be kept. Discuss why observation and interpretation must be recorded separately, with examples.
7. Discuss the ethics of field observation and interviewing. Explain what informed and continuing consent requires, and identify the groups that need particular care.
8. Explain the interview funnel. Contrast four pairs of good and bad interview questions and explain in each case why the poor question produces unusable data.
9. Explain laddering and the five whys with a worked chain of your own. Distinguish stated, observed and latent needs, and explain why latent needs carry the design opportunity.
10. Describe the process of synthesis from field notes to insight. Distinguish an observation from an insight and explain why each cluster must be named with a sentence rather than a word.
11. Discuss the three ways a problem can be badly framed, with examples. State the anatomy of a good problem statement and explain the function of each component.
12. Explain How Might We questions. Discuss how a question can be too tight or too loose, and write six questions of your own from a single problem statement.
13. Explain the point-of-view template and the four tests of a finished statement. Discuss why needs must be verbs, with three examples rewritten from noun form.
14. Explain problem reframing and describe the four reframing moves. Illustrate with at least two case studies from Unit I, identifying the reframing in each.

Text: Tim Brown, *Change by Design*, Harper Collins, 2012 — Chapters 3 and 4; Idris Mootee, *Design Thinking for Strategic Innovation*, Wiley, 2013 — Chapters 4 and 5. **References:** Jeanne Liedtka and Tim Ogilvie, *Designing for Growth*, Columbia Business School Publishing, 2011 — the chapters on What Is, journey mapping and value-chain analysis; Nigel Cross, *Engineering Design Methods*, 4th Edn., Wiley, 2008 — the chapters on clarifying objectives and establishing functions.
